

Native FeynCalc Amplitudes

Process: e- + gamma -> e- + gamma

Backend: feynarts_feyncalc_native

Expression source: feynncalc_amplitudes.m

Diagram 1: s-channel

$$\mathcal{M}_s = \frac{i\bar{\varepsilon}^{\text{Lor1}}(k1)\bar{\varepsilon}^{*\text{Lor2}}(k2) \left(\varphi(\overline{p2}, m_e)\right) \cdot \left(ie\bar{\gamma}^{\text{Lor2}}\right) \cdot \left(\bar{\gamma} \cdot (\overline{k2} + \overline{p2}) + m_e\right) \cdot \left(ie\bar{\gamma}^{\text{Lor1}}\right) \cdot \left(\varphi(\overline{p1}, m_e)\right)}{(-\overline{k2} - \overline{p2}) - m_e^2}$$

Diagram 2: u-channel

$$\mathcal{M}_u = \frac{i\bar{\varepsilon}^{\text{Lor1}}(k1)\bar{\varepsilon}^{*\text{Lor2}}(k2) \left(\varphi(\overline{p2}, m_e)\right) \cdot \left(ie\bar{\gamma}^{\text{Lor1}}\right) \cdot \left(\bar{\gamma} \cdot (\overline{p2} - \overline{k1}) + m_e\right) \cdot \left(ie\bar{\gamma}^{\text{Lor2}}\right) \cdot \left(\varphi(\overline{p1}, m_e)\right)}{(\overline{k1} - \overline{p2}) - m_e^2}$$

Total amplitude

$$\mathcal{M}_{\text{total}} = \frac{i\bar{\varepsilon}^{\text{Lor1}}(k1)\bar{\varepsilon}^{*\text{Lor2}}(k2) \left(\varphi(\overline{p2}, m_e)\right) \cdot \left(ie\bar{\gamma}^{\text{Lor1}}\right) \cdot \left(\bar{\gamma} \cdot (\overline{p2} - \overline{k1}) + m_e\right) \cdot \left(ie\bar{\gamma}^{\text{Lor2}}\right) \cdot \left(\varphi(\overline{p1}, m_e)\right)}{(\overline{k1} - \overline{p2}) - m_e^2} + \frac{i\bar{\varepsilon}^{\text{Lor1}}(k1)\bar{\varepsilon}^{*\text{Lor2}}(k2) \left(\varphi(\overline{p2}, m_e)\right) \cdot \left(ie\bar{\gamma}^{\text{Lor2}}\right) \cdot \left(\bar{\gamma} \cdot (\overline{k2} + \overline{p2}) + m_e\right) \cdot \left(ie\bar{\gamma}^{\text{Lor1}}\right) \cdot \left(\varphi(\overline{p1}, m_e)\right)}{(-\overline{k2} - \overline{p2}) - m_e^2}$$